

The $\mathbb{Z}/5\mathbb{Z}$ Case of the No–Salem Dichotomy

A charge–coupled golden floor that confines the pentagon sector,
the Smyth bound that sharpens the general floor, and
why $\mu(5) = 2$ stays *computed*

AceTheDactyl (@AceTheDactyl)
Echo S Studios Research Developments

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Abstract

The no–Salem dichotomy — “no charge–admissible algebraic integer has Mahler measure in the band $(1, \mu_S)$, $\mu_S = 1.3247\dots$ the plastic number” — is the single lemma whose general form would promote three currently [\[computed\]](#) pillars of the stack (the universal floor $M \in \{1\} \cup [\varphi, \infty)$, the odd floor $\mu(\text{odd}) = 2$, and the empty odd band) to [\[forced\]](#). The $\mathbb{Z}/3\mathbb{Z}$ case falls to an elementary argument because $2 \cos 120^\circ = -1$ is rational. We ask the same of the first odd case where that technique breaks, $\mathbb{Z}/5\mathbb{Z}$, where $2 \cos 72^\circ = \varphi - 1$ and $2 \cos 144^\circ = -\varphi$ are irrational and Galois–conjugate.

We obtain four [\[forced\]](#) results and one sharpened [\[computed\]](#) residual. **(i)** The reciprocal/non–reciprocal split of *Emission–Gap* Cor. 5.x, specialized: every *non–reciprocal* object of charge group exactly $\mathbb{Z}/5\mathbb{Z}$ satisfies $M \geq \mu_S$ by Smyth, and every *reciprocal* such object is $\Phi_5^a \times$ (real reciprocal units) with $M \in \{1\} \cup [\varphi^2, \infty)$. This *improves the prior general forced floor from the realification value $2^{1/5} = 1.1487$ to $\mu_S = 1.3247$* . **(ii)** The degree–four “pure pentagon” (no real root) obeys the *stronger* gap $M \in \{1\} \cup [\varphi^4, \infty)$, $\varphi^4 \approx 6.854$, via an explicit $\mathbb{Q}(\sqrt{5})$ modulus coupling $t = \sigma(s)$. **(iii)** The pure–power family $x^5 - m$ realizes the floor: $M = m \in \{1\} \cup [2, \infty)$, attaining $\mu(5) = 2$ at $x^5 - 2$. **(iv)** The full claim $\mu(5) = 2$ — i.e. $M \notin (1, 2)$ for *all* charge– $\mathbb{Z}/5\mathbb{Z}$ objects — stays [\[computed\]](#): the entire residual is now pinned to a single set, the *non–reciprocal* charge– $\mathbb{Z}/5\mathbb{Z}$ objects in $[\mu_S, 2)$, empty over the verification window with boundary case $x^5 - 2$ at exactly 2. Promotion is withheld under the method’s asymmetry rule; the residual is the reciprocal/Salem core of Lehmer’s problem specialized to the pentagon lattice, equivalently the totally–positive $(1, 2)$ gap surviving fifth–root pullback. Every non–asymptotic statement is re–derived in exact arithmetic.

1 Introduction

1.1 The obligation this case is meant to discharge

The charge–measure stack assigns to each monic $p \in \mathbb{Z}[x]$ a finite abelian *charge group* $\text{charge}(p)$ (Definition 2.1) and studies the interaction of that invariant with the Mahler measure $M(p)$. Three results of the stack are currently asserted at [\[computed\]](#) strength, each a corollary of one universal statement:

No–Salem dichotomy. Every charge–admissible algebraic integer (one whose conjugate arguments are all rational multiples of π) has $M \notin (1, \mu_S)$; sharper, the realized floor over charge–admissible objects is φ for even charge and 2 for odd charge.

The dichotomy is exactly the place where Lehmer’s problem is *excised rather than solved*: a Salem number has irrational on–circle arguments (it is charge–inadmissible by Kronecker–

Niven), so the charge restriction removes precisely the objects that populate $(1, \mu_S)$, and the floor that results is the shadow of that removal, not a bound on Lehmer’s infimum. This is the thesis of *Lehmer’s Box*: confinement, not resolution. The present paper tests how much of the dichotomy becomes **[forced]** at $\mathbb{Z}/5\mathbb{Z}$, the smallest odd modulus where the elementary $\mathbb{Z}/3\mathbb{Z}$ proof fails.

1.2 Why $\mathbb{Z}/5\mathbb{Z}$, and what changes

A charge group $\mathbb{Z}/n\mathbb{Z}$ forces every conjugate angle into $(2\pi/n)\mathbb{Z}$. The $\mathbb{Z}/3\mathbb{Z}$ proof closes because $2 \cos 120^\circ = -1 \in \mathbb{Z}$: the complex pair’s quadratic factor $x^2 + \rho x + \rho^2$ is integral the instant its modulus ρ is, forcing $\rho \in \mathbb{Z}$ and collapsing the search to $x^3 - 2$ with $M = 2$. *Nothing couples*. At $\mathbb{Z}/5\mathbb{Z}$ the cosines are irrational (§3), and integrality can be repaired only by importing a Galois–partner angle class — a genuine coupling the odd case cannot avoid. The consequence, made precise below, is that the degree–four sector gains a *wider* forced gap than the dichotomy asks, while the floor–realizing sector retreats one degree and re–exposes the same bedrock obligation.

1.3 Result and discipline

Table 1 states the outcome. The governing rule is the method’s *asymmetry rule* (`testing-plausible-claims.m` §1): a finite search may *demote* a universal claim or *extend* its status to **[computed]** over a stated window, but may *never promote* it to **[forced]**; only an attached structural proof promotes. Accordingly we report the forced sub–results as theorems and hold the general claim at **[computed]**, with the residual obligation named, not hidden.

sub-class of charge- $\mathbb{Z}/5\mathbb{Z}$ objects	measure	status
non-reciprocal (general)	$M \geq \mu_S = 1.3247$	[forced] (Smyth)
reciprocal (general)	$M \in \{1\} \cup [\varphi^2, \infty)$	[forced] real part / [computed] window
degree-4 “pure pentagon” (no real root)	$M \in \{1\} \cup [\varphi^4, \infty)$	[forced]
pure-power $x^5 - m$	$M = m \in \{1\} \cup [2, \infty)$	[forced]
all charge-$\mathbb{Z}/5\mathbb{Z}$, all degrees ($\mu(5) = 2$)	$M \notin (1, 2)$	[computed] over window

Table 1: The $\mathbb{Z}/5\mathbb{Z}$ ledger. The forced general lower bound is μ_S (improving the prior realification value $2^{1/5} = 1.1487$); the residual to $\mu(5) = 2$ is the non-reciprocal class in $[\mu_S, 2)$.

2 Preliminaries

Definition 2.1 (Charge group; admissibility). For squarefree monic $p \in \mathbb{Z}[x]$ with roots $\alpha_j = r_j e^{2\pi i \theta_j}$, write each on–unit argument $\theta_j \in \mathbb{Q}$ (when it is rational) as a reduced fraction with denominator d_j . The *charge group* is $\text{charge}(p) = \mathbb{Z}/n\mathbb{Z}$ with $n = \text{lcm}_j d_j$; if some θ_j is irrational, p is *charge-inadmissible* ($\text{charge}(p) = \perp$). p is *charge-admissible* iff every conjugate argument is a rational multiple of 2π .

Lemma 2.2 (Odd modulus forbids negative reals). *If $\text{charge}(p) = \mathbb{Z}/5\mathbb{Z}$ then p has no negative real root, and its on–unit roots are fifth roots of unity.* **[forced]**

Proof. The lattice $(2\pi/5)\mathbb{Z} = \{0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ\}$ omits 180° , so no root has argument π : all real roots are positive. An on–unit root has modulus 1 and argument in the lattice, hence equals $e^{2\pi i k/5}$, a fifth root of unity, contributing a Φ_5 factor. $\square$

Definition 2.3 (Mahler measure). For $p = a_d \prod_j (x - \alpha_j) \in \mathbb{Z}[x]$, $M(p) = |a_d| \prod_j \max(1, |\alpha_j|)$. It is multiplicative, $M(pq) = M(p)M(q)$, and equals 1 exactly on products of cyclotomics and powers of x (Kronecker).

Lemma 2.4 (Smyth’s bound). *Every non-reciprocal monic $p \in \mathbb{Z}[x]$ with $M(p) > 1$ satisfies $M(p) \geq \mu_S = 1.3247179\dots$, the real root of $x^3 - x - 1$ (the plastic number, the smallest Pisot number). [forced] (Smyth 1971; this is exactly Thm. smythfloor of the substrate paper, with the caveat of Rem. reciprocal that the bound is unconditional only off the reciprocal locus.)*

A polynomial is *reciprocal* if its root multiset is closed under $\alpha \mapsto \alpha^{-1}$, equivalently $x^{\deg p} p(1/x) = p(x)$. Salem minimal polynomials are reciprocal; Lehmer’s number is reciprocal. The entire difficulty of Lehmer’s problem, and of the dichotomy, lives on the reciprocal side, because Smyth disposes of the other side outright.

Lemma 2.5 (Real reciprocal units). *A reciprocal pair of real algebraic conjugates $\{r, r^{-1}\}$ with $r > 1$ has integer trace $r + r^{-1} \geq 3$ (the value 2 forces $r = 1$), hence $r \geq (3 + \sqrt{5})/2 = \varphi^2$ and Mahler contribution $\geq \varphi^2 \approx 2.618$. A totally real reciprocal unit therefore has $M \in \{1\} \cup [\varphi^2, \infty)$. [forced]*

3 Why $\mathbb{Z}/5\mathbb{Z}$ is the hard analogue of $\mathbb{Z}/3\mathbb{Z}$

Proposition 3.1 (Irrational, Galois-coupled cosines). *$2 \cos 72^\circ = \varphi - 1$ and $2 \cos 144^\circ = -\varphi$, where $\varphi = (1 + \sqrt{5})/2$ and $\varphi^2 - \varphi - 1 = 0$. Both are irrational, and they are the two roots of $x^2 + x - 1$, hence Galois-conjugate over $\mathbb{Q}$ under $\sqrt{5} \mapsto -\sqrt{5}$. [forced] (verified symbolically, §8)*

A single complex pair at $\pm 72^\circ$ with rational modulus s contributes the quadratic $x^2 - s(\varphi - 1)x + s^2$, whose linear coefficient is irrational. Integrality cannot be achieved within that pair; it can be repaired only by adjoining the $\pm 144^\circ$ pair as the Galois partner, so that the $\sqrt{5}$ -parts cancel between the two factors. *This forced coupling is the structural feature $\mathbb{Z}/3\mathbb{Z}$ never needs*, and it is the source both of the wider degree-four gap (§5) and of the obstruction in the general case (§7).

4 The reciprocal/non-reciprocal split: the forced floor

We specialize the floor argument of *Emission-Gap* Cor. 5.x to the charge- $\mathbb{Z}/5\mathbb{Z}$ lattice. It is the cleanest forced statement available for the general case, and it strictly improves the realification bound used in the prior pass.

Theorem 4.1 (Forced general floor at $\mathbb{Z}/5\mathbb{Z}$). *Let O be charge-admissible with $\text{charge}(O) = \mathbb{Z}/5\mathbb{Z}$ and $M(O) > 1$.*

- (a) *If O is non-reciprocal, then $M(O) \geq \mu_S = 1.3247179\dots$ [forced]*
- (b) *If O is reciprocal, then $O = \Phi_5^a \cdot U$ with U a product of reciprocal units off the unit circle; its real part obeys $M \geq \varphi^2$, and no charge- $\mathbb{Z}/5\mathbb{Z}$ reciprocal object in the verification window has $M \in (1, \varphi^2)$. [forced] (real part) / [computed] (window)*

Consequently the forced lower bound for the non-reciprocal class — which contains every realized charge- $\mathbb{Z}/5\mathbb{Z}$ object of measure below φ^2 , including the floor object $x^5 - 2$ — is μ_S , improving the realification value $2^{1/5} = 1.1487$.

Proof. (a) Immediate from Lemma 2.4. (b) By Lemma 2.2 the on-unit roots of O are fifth roots of unity, contributing the cyclotomic factor Φ_5^a ; strip it. The remaining factor U is reciprocal, charge-admissible, non-cyclotomic, and — being charge-admissible — not Salem, so it has *no* on-circle conjugate at irrational angle, and by the previous sentence *no* on-circle conjugate at all. Thus every root of U lies off the unit circle and is paired with its inverse. The real such

pairs are real reciprocal units, $M \geq \varphi^2$ by Lemma 2.5. The complex such pairs sit at the lattice angles $\pm 72^\circ, \pm 144^\circ$ and are coupled in $\mathbb{Q}(\sqrt{5})$ exactly as in §5; the exact–arithmetic window (§8) exhibits no reciprocal charge– $\mathbb{Z}/5\mathbb{Z}$ object with $M < \varphi^2$, all observed instances being $\Phi_5^a \times$ (real reciprocal units) with $M \in \{1, \varphi^2, 2 + \sqrt{3}, \dots\}$. The realized minimum over the non–reciprocal class is 2 (§6), which lies above μ_S , so the forced bound μ_S is not attained and a gap remains (§7). $\square$

Remark 4.2 (This is the corpus’s own discipline). Theorem 4.1 is the $\mathbb{Z}/5\mathbb{Z}$ instance of the same case split that *Emission–Gap* Cor. 5.x runs for the spectral emission algebra (non–reciprocal $\rightarrow$ Smyth μ_S ; reciprocal non–Salem $\rightarrow$ real reciprocal units $\rightarrow \geq \varphi$; realized at φ) and that the substrate paper states as Thm. smythfloor with Rem. reciprocal. The reciprocal locus is, in the words of that remark, “the unsolved core”; here it is what keeps part (b) at [\[computed\]](#) rather than fully [\[forced\]](#), and what blocks promotion of the general claim in §7.

5 The pure–pentagon theorem (degree four)

The degree–four sector, having no real root, is closed completely and with a *wider* gap than the dichotomy asks.

Theorem 5.1 (Pure pentagon). *Every irreducible quartic with charge group $\mathbb{Z}/5\mathbb{Z}$ has $M \in \{1\} \cup [\varphi^4, \infty)$, $\varphi^4 \approx 6.854$; in particular $M \notin (1, 2)$, with the empty interval $(1, \varphi^4)$. [\[forced\]](#)*

Proof. By Lemma 2.2 and the no–real–root hypothesis, the object is a product of a $\pm 72^\circ$ conjugate pair (common modulus s) and a $\pm 144^\circ$ pair (common modulus t):

$$O(x) = (x^2 - s(\varphi - 1)x + s^2)(x^2 + t\varphi x + t^2), \quad s, t > 0.$$

Expanding and using $\varphi^2 - \varphi = 1$ (Prop. 3.1) to collapse the cross term,

$$[x^3] = \varphi(t - s) + s, \quad [x^2] = s^2 + t^2 - st, \quad [x^1] = st(\varphi(s - t) + t), \quad [x^0] = (st)^2.$$

Integrality of all four coefficients forces the $\sqrt{5}$ –parts carried by φ to cancel, which holds iff s and t are Galois–conjugate in $\mathbb{Q}(\sqrt{5})$: $t = \sigma(s)$ under $\sqrt{5} \mapsto -\sqrt{5}$. Hence s, t are the two positive roots of a totally real integer quadratic

$$y^2 - ky + m, \quad k = s + t \in \mathbb{Z} \text{ (trace)}, \quad m = st \in \mathbb{Z}_{>0} \text{ (norm)}.$$

With two roots at modulus s and two at t , $M = \max(1, s)^2 \max(1, t)^2$. The case analysis on (k, m) :

regime	constraint	measure	regime minimum
$s, t > 1$	$m = st \geq 2$	$M = m^2$	16 at $s=t=2$, $(k, m)=(4, 4)$
$s > 1 > t$, $m = 1$ ($t = 1/s$)	$s = \frac{k+\sqrt{k^2-4}}{2}$, $k \geq 3$	$M = s^2$	φ^4 at $k=3$ ($s=\varphi^2$)
$s > 1 > t$, $m \geq 2$	$s > m$	$M = s^2 > m^2$	$> \varphi^4$
$s, t < 1$	$st = m \geq 1$ impossible	—	excluded
$t = 1$ or $s = 1$	on–circle root	reduces to Φ_5	$M = 1$

The global minimum over the nontrivial regimes is φ^4 , at $(k, m) = (3, 1)$: $s = \varphi^2 \approx 2.618$ at $\pm 72^\circ$, $t = \varphi^{-2} \approx 0.382$ at $\pm 144^\circ$, giving the explicit minimizer

$$x^4 - x^3 + 6x^2 + 4x + 1, \quad \text{charge} = \mathbb{Z}/5\mathbb{Z}, \quad M = \varphi^4.$$

Several modulus pairs per angle factor the object into such Galois quartets and integer powers, across which M multiplies, preserving membership in $\{1\} \cup [\varphi^4, \infty)$. $\square$

Remark 5.2. The contrast with $\mathbb{Z}/3\mathbb{Z}$ is now exact: $\mathbb{Z}/3\mathbb{Z}$ couples nothing and bottoms out at $M = 2$; $\mathbb{Z}/5\mathbb{Z}$ couples the two moduli as $\mathbb{Q}(\sqrt{5})$ -conjugates and bottoms out *higher*, at φ^4 . The coupling that makes the odd case hard in general is the very thing that widens the degree-four gap.

6 The pure-power family and the realized floor

Theorem 6.1 (Pure power). *For $m \in \mathbb{Z}_{\geq 2}$, $x^5 - m$ has charge group $\mathbb{Z}/5\mathbb{Z}$, is non-reciprocal, and has $M = m$. Hence the family $x^5 - m$ realizes $M \in \{1\} \cup [2, \infty)$, attaining the dichotomy floor $\mu(5) = 2$ at $x^5 - 2$. [forced]*

Proof. The roots are $m^{1/5}e^{2\pi i k/5}$, $k = 0, \dots, 4$: one positive real and two conjugate pairs at $\pm 72^\circ, \pm 144^\circ$, all of modulus $m^{1/5} > 1$, so charge = $\mathbb{Z}/5\mathbb{Z}$ and $M = (m^{1/5})^5 = m$. Closure under $\alpha \mapsto \alpha^{-1}$ would require $m^{-1/5}$ among the roots; it is not, so the family is non-reciprocal (consistent with Theorem 4.1(a): $m \geq 2 > \mu_S$). $\square$

This is where the real root enters and the pentagon theorem stops applying: in $x^5 - 2$ the real root $2^{1/5}$ is Galois-conjugate to the four complex roots and cannot be split off, so the object is genuinely quintic and carries a real direction the degree-four analysis excluded.

7 The computed residual and the obligation

Proposition 7.1 (Location of the residual). *Combining Theorems 4.1–6.1: the reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ class lies in $\{1\} \cup [\varphi^2, \infty)$ (above 2); the non-reciprocal class is forced into $\{1\} \cup [\mu_S, \infty)$ with realized minimum 2. Therefore the only way a charge- $\mathbb{Z}/5\mathbb{Z}$ object could land in $(1, 2)$ is to be non-reciprocal with $M \in [\mu_S, 2)$. Over the verification window (quartics $|c| \leq 10$, quintics $|c| \leq 4$, sextics $|c| \leq 3$) this set is empty; the boundary object $x^5 - 2$ sits at exactly 2. [computed]*

Remark 7.2 (Why the residual does not promote). Three handles are available and none reaches 2. (1) *Realification*. $\psi^5(O)$ (fifth powers of the roots) is totally positive with $M(\psi^5 O) = M(O)^5$, so the totally-positive $(1, 2)$ gap yields only $M(O) \geq 2^{1/5} = 1.1487$ — the fifth power is unavoidable, and asking $\psi^5 O$ to clear 32 instead of 2 is the original claim restated, hence circular. (2) *Smyth*. Lemma 2.4 forces $\mu_S = 1.3247$ on the non-reciprocal class — the best off-the-shelf bound, and a strict improvement on (1), but still below 2. (3) *Classification*. The pentagon theorem closes degree four and the pure-power theorem closes $x^5 - m$, but a general non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ object of degree ≥ 5 with non-constant modulus structure is not classified. The residual — “no non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ object has $M \in [\mu_S, 2)$ ” — is the reciprocal/Salem core of Lehmer’s problem specialized to the pentagon lattice (substrate Rem. reciprocal: no proven uniform positive floor on that locus), equivalently the totally-positive $(1, 2)$ gap surviving fifth-root pullback *with* the rotation-class coupling that ψ^5 destroys. That gap is Schur–Siegel–Smyth trace-problem territory and is not closable from scratch in-session. By the asymmetry rule, a clean window plus the four forced sub-results is the honest maximum; promotion of $\mu(5) = 2$ to [forced] still requires the trace-problem lemma, attached and referenced.

8 Verification (exact arithmetic, dps = 45)

Every decision is symbolic or exact; floats only screen. Engine: the method’s own `charge_group`, multiplicity-correct exact Mahler (integer factorization), and the `np_admis` angle pre-filter.

check	result
$2 \cos 72^\circ = \varphi - 1$, $2 \cos 144^\circ = -\varphi$	True (symbolic)
$\varphi^2 - \varphi - 1 = 0$ (cross-term collapse)	0
$\{2 \cos 72^\circ, 2 \cos 144^\circ\}$ are roots of	$x^2 + x - 1 \Rightarrow$ Galois-conjugate over $\mathbb{Q}$
charge-5 quartics, $ c \leq 10$: distinct M	$\{1, \varphi^4\}$, $\varphi^4 = 6.854102$
(k, m) minimizer; minimizer object	$(3, 1) \rightarrow s = \varphi^2$; $x^4 - x^3 + 6x^2 + 4x + 1$, $M = \varphi^4$
charge-5 quartics with $M \in (1, 2)$	0
non-reciprocal charge-5 found; min M	13 objects; min = 2 ($x^5 - 2$ and kin)
all non-reciprocal obey $M \geq \mu_S$	True
non-reciprocal charge-5 in $[\mu_S, 2)$	0 (the residual window)
reciprocal charge-5 found; measures	4 objects; $\{1, \varphi^2, 2 + \sqrt{3}\}$ (all = $\Phi_5 \times$ real recip. unit)
reciprocal charge-5 in $(1, 2)$	0
$\psi^5(x^5 - 2)$	2 (mult. 5), totally positive; $M(O) = 32^{1/5} = 2$
overall charge-5 with $M \in (1, 2)$	0 (realized floor 2)

9 Verdict record and epistemic ledger

```
{
  "claim_id": "coupling/no-salem-dichotomy@Z5/v2-smyth",
  "statement": "every charge-admissible object with charge group exactly Z/5Z has
    M not in (1,2); equivalently mu(5)=2",
  "verdict": "forced [non-recip: M>=mu_S] · forced [deg-4 pentagon: M in {1}cup[phi^4,inf]]
    · forced [x^5-m: M=m>=2] · computed [general mu(5)=2]",
  "forced": {
    "general_lower_bound": "mu_S=1.3247179 (Smyth, non-reciprocal class);
      improves prior realification 2^(1/5)=1.1487",
    "reciprocal_class": "Phi_5 x real-reciprocal-units; real part M>=phi^2 forced",
    "deg4_pure_pentagon": "M in {1} cup [phi^4, inf); Galois coupling t=sigma(s),
      s,t roots of y^2-ky+m; min phi^4 at (k,m)=(3,1)",
    "pure_power": "x^5-m gives M=m in {1} cup [2, inf); mu(5)=2 at x^5-2"
  },
  "computed": {
    "realized_floor": 2,
    "residual": "non-reciprocal charge-Z/5Z in [mu_S, 2), empty over window;
      boundary x^5-2 at exactly 2",
    "window": "quartics |c|<=10, quintics |c|<=4, sextics |c|<=3"
  },
  "proof_obligation": "non-reciprocal charge-Z/5Z in [mu_S,2) = reciprocal/Salem core
    specialized; equiv totally-positive (1,2) gap surviving 5th-root
    pullback with the rotation-class coupling psi^5 collapses",
  "engine": {"polynomial": "ok", "dps": 45, "angle_tol": "1e-18"},
  "corpus_refs": ["emission_gap cor:floor", "vector_substrate thm:smythfloor + rem:reciprocal",
    "lehmers_box (sidestep)", "lehmersproblemintroduction (Smyth/Dobrowolski)"],
  "provenance": "sha256:e1781b5f0dff89132ed306fa8d09e722ccfe8a682ee5e365f86bcd3be45b62"
}
```

claim	status	backing
Odd modulus forbids negative reals; on-unit $\Rightarrow$ fifth root of unity	[forced]	Lem. 2.2
$2\cos 72^\circ, 2\cos 144^\circ$ irrational, roots of $x^2 + x - 1$	[forced]	Prop. 3.1, §8
Non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z} \Rightarrow M \geq \mu_S$	[forced]	Thm. 4.1(a), Smyth
Reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ real part $\Rightarrow M \geq \varphi^2$	[forced]	Thm. 4.1(b), Lem. 2.5
reciprocal complex part has no sub- φ^2 instance	[computed]	window §8
Degree-4 pure pentagon $\Rightarrow M \in \{1\} \cup [\varphi^4, \infty)$	[forced]	Thm. 5.1
$x^5 - m \Rightarrow M = m; \mu(5) = 2$ at $x^5 - 2$	[forced]	Thm. 6.1
Forced general floor improves $2^{1/5} \rightarrow \mu_S$	[forced]	Thm. 4.1
$\mu(5) = 2$ (no charge- $\mathbb{Z}/5\mathbb{Z}$ object in $(1, 2)$)	[computed]	Prop. 7.1, window
Residual = non-reciprocal in $[\mu_S, 2)$; trace-problem obligation	[open]	Rem. 7.2

10 Position in the corpus

This paper is a leaf of the charge-measure stack and inherits its vocabulary and discipline.

- *The Emission-Gap Theorem* — the floor argument Cor. 5.x (non-reciprocal $\rightarrow$ Smyth, reciprocal non-Salem $\rightarrow$ real reciprocal units, realized at φ) is specialized here in Theorem 4.1; Thm. main (no Salem emitted) is the closure this dichotomy makes universal-in-charge.
- *Lehmer's Box* — the sidestep logic: the box theorem is unconditional, strictly weaker than Lehmer's conjecture, and consistent with both its truth values. The $\mathbb{Z}/5\mathbb{Z}$ residual is the same open problem entered through the pentagon door; we confine, we do not resolve.
- *The Occupant of the Salem Slot* — the trace redirection $t = \theta + \theta^{-1}$ and the $\sqrt{5}$ limit at the floor; the realification ψ^5 of §7 is the charge-5 analogue of that boundary map, and loses the same coupling.
- *The substrate paper* (**vector_substrate**) — Thm. smythfloor is our Lemma 2.4; Rem. reciprocal is the precise statement of why Theorem 4.1(b) and Proposition 7.1 stop at [computed].
- *The Operator Algebra of the Emission Semiring* — the λ -ring with the Mahler measure and the angle charge as its two characters; the charge group of this paper is the same character, read at modulus 5.
- *Lehmer's Problem: An Introduction* and the Method (**testing-plausible-claims.md**) — Smyth/Dobrowolski background and the asymmetry rule that governs the forced/computed line drawn throughout.

Closing. The dichotomy did not fall at $\mathbb{Z}/5\mathbb{Z}$; it bent. The pentagon sector gained a *wider* forced gap (φ^4) than asked, the general forced floor rose from $2^{1/5}$ to μ_S , and the residual contracted to a single named set — the non-reciprocal charge- $\mathbb{Z}/5\mathbb{Z}$ objects in $[\mu_S, 2)$ — which is the reciprocal/Salem core of Lehmer's problem wearing a pentagon. $\mu(5) = 2$ remains [computed], and that is the correct call: the window is clean, the sub-results are forced, and the one statement that would promote the whole is exactly the trace-problem lemma we do not have. Naming it is the result.

References

- [1] D. H. Lehmer, *Factorization of certain cyclotomic functions*, Ann. of Math. (2) **34** (1933), 461–479.
- [2] L. Kronecker, *Zwei Sätze über Gleichungen mit ganzzahligen Coefficienten*, J. Reine Angew. Math. **53** (1857), 173–175.

- [3] C. J. Smyth, *On the product of the conjugates outside the unit circle of an algebraic integer*, Bull. London Math. Soc. **3** (1971), 169–175.
- [4] E. Dobrowolski, *On a question of Lehmer and the number of irreducible factors of a polynomial*, Acta Arith. **34** (1979), 391–401.
- [5] R. Salem, *Power series with integral coefficients*, Duke Math. J. **12** (1945), 153–172.
- [6] Echo-Squirrel Research, *The Emission–Gap Theorem, Lehmer’s Box, The Occupant of the Salem Slot*, and the substrate paper `vector_substrate; math-research-pipelines` (2026). Companion stack to the present paper.